

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Pharm. (New) Examination University Theory Examination October / November 2016

PH221 - Pharmacology-II

Date: 21/10/2016, Friday

Time: 1.30 PM to 4.30 PM

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
 2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
 3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
 4. Figures to right indicate marks.
 5. Draw neat sketches wherever necessary.
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SECTION - I

Q 1 Attempt all questions. Each question is of one mark.

20

1. Irreversible cell injury is associated with
 - A. Clumping of chromatin
 - B. Endoplasmic swelling
 - C. Reduced oxidative phosphorylation
 - D. Profound membrane disturbance
2. Regarding hyperplasia which statement is correct?
 - A. It is never seen in the same tissue as hypertrophy
 - B. it is seen in cardiac muscle in hypoxic patients
 - C. it is limited to cells capable of mitotic division
 - D. it is rarely physiologic
3. Ischaemic injury
 - A. May be caused by any state of reduced oxygen delivery
 - B. Usually results in a necrotic pattern of injury
 - C. Leads to reduced glycolysis in the cell
 - D. Is associated with an efflux of calcium and sodium
4. Reversible cell injury is characterised by which morphological changes?
 - A. Membrane blebbing
 - B. Detachment of ribosomes from the endoplasmic reticulum
 - C. Clumping of nuclear chromatin
 - D. All of above
5. What is pyknosis?
 - A. Nuclear Condensation
 - B. Nuclear Fragmentation
 - C. Nuclear Dissolution
 - D. Hypertrophy
6. One of the following is an example of antacid
 - A. Omeprazole
 - B. Aluminium hydroxide gel
 - C. Levodopa
 - D. Diazepam

7. Example(s) of Bulk laxatives consist of
 - A. Methyl cellulose
 - B. Steculia
 - C. Ispaghula
 - D. All of above
8. What are the drugs included in Triple therapy, used for H-pylori infection?
 - A. Antiemetics, Proton pump inhibitor & amoxicillin
 - B. Proton pump inhibitor, amoxicilin and Metronidazole
 - C. Proton pump inhibitor, Antacid and Penicillin
 - D. Purgatives, Antacid and Penicillin
9. Which of the following is not a Stimulant Laxative?
 - A. Bisacodyl
 - B. Glycerol
 - C. Docusate Sodium
 - D. Sodium picosulfate
10. Example(s) of "protective factor(s)" in peptic ulcer disease
 - A. gastric mucus
 - B. prostaglandins
 - C. bicarbonate
 - D. All of above
11. Mucokinetic is a drug which:
 - A. Reduces airway mucus secretion
 - B. Increases airway mucus secretion
 - C. Makes respiratory secretions more watery
 - D. Stimulates mucociliary activity of bronchial epithelium
12. The following antiasthma drug is not a bronchodilator
 - A. Ipratropium bromide
 - B. Theophylline
 - C. Formoterol
 - D. Sodium cromoglycate
13. Select the correct statement about salmeterol
 - A. It is a long acting selective β_2 agonist bronchodilator
 - B. It is a bronchodilator with anti-inflamma-tory property
 - C. It is a β blocker that can be safely given to asthmatics
 - D. It is an antihistaminic with mast cell stabi-lizing property
14. Methylxanthines exert the following action(s) at cellular/molecular level:
 - A. Intracellular release of Ca^{2+}
 - B. Antagonism of adenosine
 - C. Inhibition of phosphodiesterase
 - D. All of the above
15. Acute Lead arsenic poisoning have symptoms of
 - A. Inflamantion of mucus membrane, Jaudice, anemia, neuropathy, liver failure
 - B. orange coloration of toungue, bone weakness, silver skin, skin irritation, immune compromised
 - C. metallic taste, GI irritation, parasthesia, muscle cramp, stool containing lead sulphate
 - D. Heart failure, sudden rise in blood pressure, kidney failure
16. Which of the following drug is antidote of metal poisoning?
 - A. Thalidomide
 - B. Sulphonamide
 - C. Loparamide
 - D. Dimercaprol

17. Full form of TNF is
 - A. Tear nephrotic factor
 - B. Tissue nephrotic factor
 - C. Tumour necrosis factor
 - D. Tissue nascent factor
18. Type IV hypersensitivity is a _____ type
 - A. Delayed
 - B. Chronic
 - C. Acute
 - D. All of above
19. Following are the different types of Prostaglandin except one
 - A. PGX2
 - B. PGE2
 - C. PGF2
 - D. PGD2
20. Following are the functional manifestation of Inflammation except
 - A. Vascular changes
 - B. Edema
 - C. Neutrophilic infiltration
 - D. Bleb formation of cell membrane

SECTION – II

- Q 2** Attempt any **TWO** of the following
- A Write a detail pathogenesis of Inflammation and indicate its manifestations 05
 - B What is the concept of Cell adaptation? Explain it in detail with examples 05
 - C Write a short note on role of oxidative stress in cell injury 05
- Q 3** Attempt any **TWO** of the following
- A Separate out reversible and irreversible cell injury by providing distinguished features of it. 05
 - B Draw schemetaic diagramme of morphological changes in cell injury culminating in Apoptosis and Necrosis. Explain Apoptosis and necrosis terms. 05
 - C Study in detail about Mercury and lead poisoning including its toxic reactions on body. 05

SECTION – III

- Q 4** Attempt any **FOUR** of the following
- A Discuss pathogenesis of Peptic ulcer. Discuss the drugs used to neutralize gastric acid secretion 05
 - B Provide mechanism, pharmacology, unwanted effects, clinical uses of Proton pump inhibitors 05
 - C Explain expectorants and antitussive drugs in brief 05

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| D | Describe pathophysiology and treatment of asthma. | 05 |
| E | Explain concept of reflex mechanism of Vomiting and discuss antiemetic drugs in brief. | 05 |
| F | Describe prostaglandins and their actions. | 05 |

Q 5 Attempt any FOUR of the following

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| A | Write a detail note on Leukotrienes | 05 |
| B | Classify histamine receptors with examples of drugs acting on it. Give therapeutic uses of anti-histaminics. | 05 |
| C | Give concept and applications of gene therapy. What is gene delivery? | 05 |
| D | Provide a brief role of Cytokines in body. | 05 |
| E | Discuss purgatives and prokinetic agents in brief. | 05 |
| F | Discuss antidiarrhoeal drug in detail. | 05 |
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